

Simultaneous improvement in strength and ductility of 3D-printed refractory high-entropy alloys by addition of molybdenum

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ABSTRACT

Molybdenum (Mo) is an important strengthening element in refractory high-entropy alloys. However, it is difficult to produce a Mo containing alloy with uniform elemental distribution because of the high melting point of Mo. This work employed 3D printing, also known as laser additive manufacturing (LAM), technique to prepare TiZrVNbAl-system refractory high-entropy alloys (RHEAs) with different Mo contents. Due to the high temperature and strong convection in melting pool, uniform elemental distribution is achieved in LAM TiZrVNbAl-system RHEAs. With increasing Mo, the solid-liquid intervals of alloys increase, which promotes the formation of fine equiaxed growth, and reducing in the mean grain sizes. Meanwhile, a large difference in the atomic radius between the Mo and the TiZrVNbAl RHEA results in strong solid-solution strengthening, and hence a 34 % increase in the yield strength. These results imply that the LAM is a promising technique to produce RHEAs with uniform elemental distribution, as well as good strength–ductility balance, which is expected to be able to widen the potential applications these refractory alloys.

1. Introduction

Refractory high-entropy alloys (RHEAs) are multi-principal elements alloys, which have been extensively studied due to excellent high temperature softening resistance and high strength [1–4]. The alloy is primarily composed of high melting point elements from IV, V and VI groups (Ti, Zr, V, Nb, W, Mo, etc.). RHEAs are known to form single-phase structures due to the high mixing entropy [1,5–8].

Lightweight TiZrVNbAl-system RHEAs are single phase BCC alloys with excellent performance, the main strengthening mechanism is solid solution strengthening [9,10]. However, the addition of Mo serves to enhance the solid solution strengthening capabilities of RHEAs, mainly due to the notable difference in atomic radius between Mo and TiZrNb-based RHEAs [11,12]. Therefore, the addition of Mo can improve the yield strength of the alloy [7,12]. Furthermore, the addition of Mo expands constitutional supercooling zones in the solid-liquid interface, leading to an increase the growth restriction factor [13,14]. The high concentrations of Mo contents often lead to a larger growth restriction factor (Q). According to Interdependence Theory, increasing the Q value can effectively promotes grain refinement [13,15]. Therefore, the addition of Mo can not only facilitate the solid solution

strengthening effect in the RHEAs, but also lead to refinement of the grain sizes.

However, RHEAs containing high melting point elements are difficult to obtain uniform distribution of elements through melting technology [2,16,17]. For example, the RHEA containing the high-melting-point Mo element, as reported by Zeng et al. while demonstrating high yield strength, is susceptible to element segregation during the casting process [12]. Laser additive manufacturing (LAM), as a bottom-up preparation technique, not only has greater design freedom and more efficient material utilization [18–21], but can also significantly reduce post-processing and shorten production cycles [22,23]. More importantly, LAM, which employs a laser as a heat source and is characterized by high melt pool temperatures and strong liquid convection, is regarded as a promising technique for preparing refractory alloys [24–27]. Xiao et al. [28] employed LAM to produce WMo-TaNbNiTi RHEAs with uniformly distributed high-melting-point elements, including W, Mo, and Ta, within the grains. Furthermore, LAM exhibits the distinctive attribute of coaxial powder feeding, which enables the concurrent preparation of different alloys through mixing powders in different proportions. This contrasts with the complex approach of synthesizing multiple alloys through the use of melting

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technology. Guo [29] et al. prepared three different types of RHEAs, designated as TiZrHfNb x ($x = 0.6, 0.8, 1.0$), through the mixing of elemental powders. The effect of varying Nb contents on the microstructure and mechanical properties of the alloys was investigated, and it was found that the yield strength of the alloys produced via LAM was superior to that of the casting process. Consequently, the utilization of LAM for the preparation of RHEAs comprising high-melting point elements enables the simultaneous fabrication of RHEAs with varying element compositions and the uniform elements distribution.

To investigate the effect of Mo content on TiZrVNbAl RHEA, this work first evenly mixed TiZrVNbAl RHEA powders and Mo elemental powders in different proportions by a centrifuge tube. The TiZrVNbAl RHEA was selected a Ti-18Zr-15Nb-12V-5Al (in at %) RHEA (named as T50 hereinafter) with excellent mechanical properties for this study [30, 31]. Research has shown that the addition of approximately 6 at % of Mo element to RHEAs can effectively improve the yield strength of alloys [11]. Therefore, this work selected Mo contents of 0, 3, 6, and 9 at % to investigate the role of Mo addition on the microstructure and mechanical properties of TiZrVNbAl RHEA. By modifying the Mo contents, a series of (T50)100-xMox ($x = 0, 3, 6, 9$ at %) RHEAs with a uniform microstructure, optimal yield strength, and enhanced ductility were successfully developed. The RHEA, which exhibits a uniform elemental distribution, was obtained through the role of the strong convection effect of the melt pool during the LAM process. The substantial atomic radius difference between the Mo element and the T50 RHEA has led to a considerable enhancement in the yield strength of alloy. The Mo6 alloy exhibits a yield strength exceeding 1 GPa and an elongation of approximately 30 %, representing a substantial enhancement over the T50 RHEA and outperforming the majority of documented RHEAs. This discovery offers a novel approach to the fabrication of RHEAs with homogeneous element distribution, optimal strength, and enhanced ductility.

2. Materials and methods

T50-RHEA spherical powders with a size range of 45 μm –150 μm were prepared by electrode induction melting gas atomization (EIGA) (Fig. 1 (a)). Mo spherical powders with a purity of 99.99 % and range of 15–53 μm , were employed (Fig. 1 (b)). The sphericity of T50 RHEA and Mo powders was determined to be 0.957 ± 0.015 and 0.901 ± 0.018 , respectively, through the utilization of dynamic image analysis. Four powders with different Mo contents were mixed in a centrifuge tube. The compositions include T50, (T50)97-Mo3, (T50)94-Mo6, and (T50)91-Mo9 RHEAs.

The T50-Mox RHEAs samples were fabricated using a 4 kW LAM system (RC-LDM4030, RAYCHAM Laser Technology Group Co., Ltd) with a 3 mm laser beam, operated in an Ar protective atmosphere with an oxygen content of less than 50 ppm. The samples were deposited onto a Ti6Al4V alloys by cyclic method, as illustrated in Fig. 2 (b). The T50-Mox RHEAs wall-like plates were prepared by the LAM system, with a

thickness of 5 mm, as depicted in Fig. 2 (c). Based on the previous research results [32,33], relevant LAM process parameters were selected: laser power: 1400 W, scanning speed: 10 mm/s, and powder feed rate ranging from 10 g/min to 12.5 g/min, a 3 mm beam diameter.

The metallographic sample is polished with SiC paper to particle size of 5000 and subsequently with SiO₂ suspension. The microstructure and static tensile fracture surface of the sample were observed using a scanning electron microscope (SEM, Regulus 8230, Hitachi, Japan). The distribution of elements was investigated utilizing an energy dispersive spectrometry (EDS). The phase structure was analyzed by X-ray diffraction (D8 Advance, Bruker AXS, Germany) using Cu K α radiation. The microstructure and grain size were analyzed using a scanning electron microscope equipped with an electron back scatter diffraction (EBSD) detector. The step size for EBSD measurements is 5 μm . The utilization of ion thinning technology for the preparation of transmission electron microscopy (TEM) samples of deformed alloys, coupled with the observation of dislocation motion during deformation through the utilization of Talos F200X equipment. Use DigitalMicrograph 4 software for TEM data processing. The phase diagram calculation method is called CALPHAD [34], which utilizes the TCHEA database integrated into Thermo Calc software. Fig. 3 shows the schematic of sampling location for XRD, SEM, EBSD testing, and tensile samples. All test samples were obtained in accordance with the XOZ plane.

A flat dog bone-shape tensile sample, with a gauge length of 15 mm, width of 1.5 mm, and thickness of 1 mm, was prepared by wire cutting in the parallel direction of the deposited sample. The sampling location and direction of the tensile sample are shown in Fig. 3. The tensile sample was performed on an AGS-X testing machine, employing a constant strain rate of $1.0 \times 10^{-3} \text{ s}^{-1}$ at room temperature. Three sample were tested for each sedimentary sample to confirm repeatability.

3. Results

3.1. Microstructure

In order to elucidate the role of changes in Mo content on the phase evolution and microstructure, XRD and back scattered electron (BSE) analyses were conducted. Fig. 4 (a) illustrated the XRD patterns of the various Mo content, with the predominant phase structure consisting of BCC. The addition of Mo did not alter the phase structure of the alloy, maintaining a stable single phase BCC structure. The diffraction peak observed at 42.5° in the Mo9 alloy is predominantly attributed to the presence of undissolved Mo particles. Fig. 4(b–e) show BSE images of different Mo content. No obvious cracks or defects are observed in the samples. The microstructure of the Mo0 alloy consists of a small amount of columnar and equiaxed grains, while the microstructures of the other three alloys are approximately equiaxed grains. Fig. 4(f) depicts the EDS spectrum of the Mo6 alloy, the elements are uniformly distributed without forming macroscopic segregation. This phenomenon is primarily attributable to the strong convection within the melt pool, which

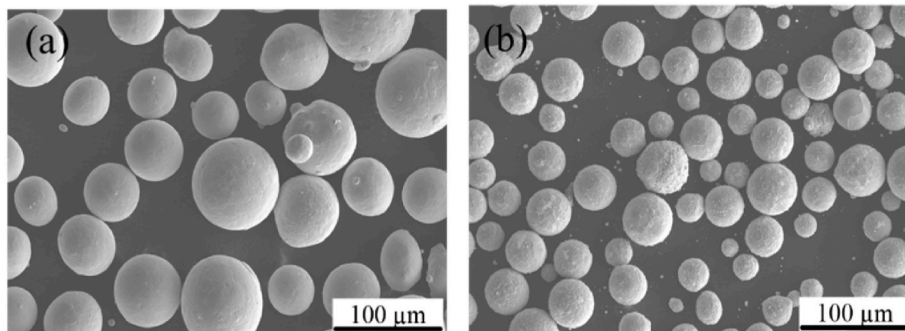


Fig. 1. Morphology of the powders observed by SEM; (a) T50-RHEA powders, (b) Mo powders.

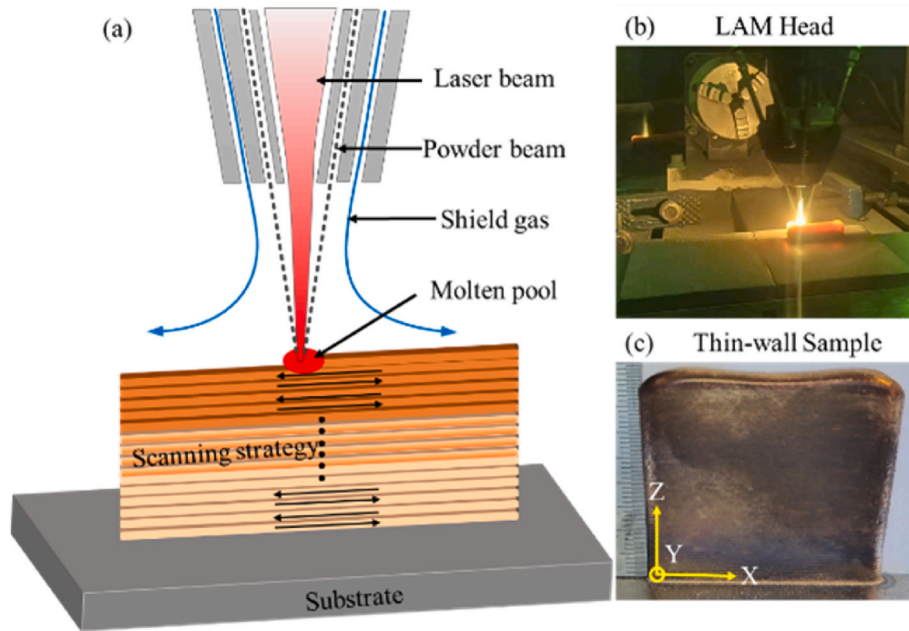


Fig. 2. (a, b) Schematic of the LAM process and fabricated bulk samples, (c) Thin-wall (T50)100-xMox RHEAs specimen fabricated by LAM.

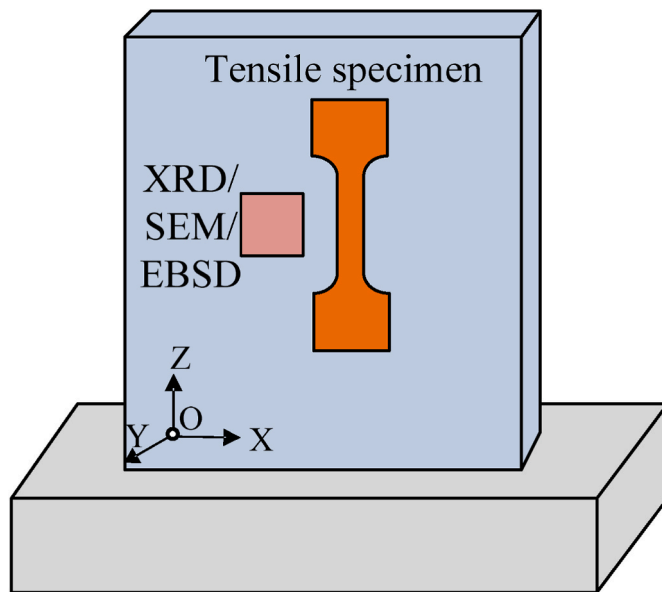


Fig. 3. Schematic of sampling location for XRD, SEM, EBSD testing, and tensile samples.

is driven by a complex interplay of thermal capillary forces, gravity, Marangoni effects, and surface tension [35,36]. An EDS analysis (Fig. S1) of Mo9 alloy indicates that the addition of excessive Mo powders will result in the formation of a minor proportion of undissolved Mo particles. As demonstrated in Table 1, EDS testing of four single-phase solid solutions reveals that the nominal composition and the actual measured values of the three alloys (Mo0, Mo3, and Mo6) are identical. In contrast, Mo in Mo9 alloy only melts at 8.01 at %, with the remaining components existing in the form of particles. This indicates that RHEAs fabricated using LAM can achieve relatively uniform elemental distribution without the requiring for solution heat treatment.

Fig. 5 depicts the inverse pole figure (IPF) and grain size distribution statistics of the as-deposited Mo0, Mo3, Mo6 and Mo9 alloys, which primarily reflect the role of Mo content on grain orientation and size. Fig. 5 (a) demonstrates that the microstructure is constituted by

columnar and equiaxed grains in the Mo0 alloy. Following the addition of Mo, the microstructure of the three alloys (Mo3, Mo6, Mo9) is predominantly constituted of equiaxed grains, with a concomitant decrease in the mean grain size, as illustrated in Fig. 5(b–d). The mean grain sizes are $183.5 \pm 3 \mu\text{m}$, $132.4 \pm 5 \mu\text{m}$, $103.2 \pm 4 \mu\text{m}$, and $83.5 \pm 3 \mu\text{m}$, respectively. In comparison with traditional alloys produced by LAM technology [37,38], these RHEAs are capable of overcoming the epitaxial growth of columnar grains during solidification, thereby forming small equiaxed grains. Furthermore, it was observed that the grain orientation was random and no strong texture was formed along the $\langle 100 \rangle$ direction, with a maximum texture strength of only 4.07, as shown in Fig. S2.

3.2. Tensile mechanical properties

To gain further insight into the role of Mo on the mechanical properties of T50 RHEA, uniaxial tensile tests were conducted at room temperature, and the results are presented in Fig. 6 (a). As the Mo content increases, there is a notable enhancement in both the yield strength and ductility. In particular, the yield strength was increased from 850 MPa for the base Mo0 alloy to 1095 MPa for the Mo6 alloy. It is noteworthy that the ductility also increased from 22 % for the base Mo0 alloy to approximately 28 % for the Mo6 alloy. This is primarily attributable to the addition of Mo element, which enhances the strain-hardening rate of Mo6 alloy (Fig. 6 (b)), prevents premature fracture, and considerably improves its ductility. Consequently, the addition of Mo not only enhances the yield strength but also improves its ductility of alloys. It should be noted that the ductility was found to decrease with an increase in Mo9 alloy. The observed reduction in elongation of the Mo9 alloy is in accordance with the findings of numerous studies on single-phase BCC structure RHEAs, predominantly due to the higher valence electron concentration (VEC) of Mo in comparison to the other constituent elements [39,40]. The addition of an excess of Mo markedly elevates the VEC of the alloy, intensifies interatomic bonding forces, and diminishes the intrinsic plasticity of the alloy [41]. Fig. 7 illustrates that the mechanical properties (yield strength, ductility) of the Mo6 alloy are superior to those of the previously reported RHEAs system.

Fig. 8 illustrates the static tensile fracture surfaces of the varying Mo content. The fracture surfaces of Mo0, Mo3, and Mo6 alloys are composed of many ductile dimples, which are typical ductile fractures

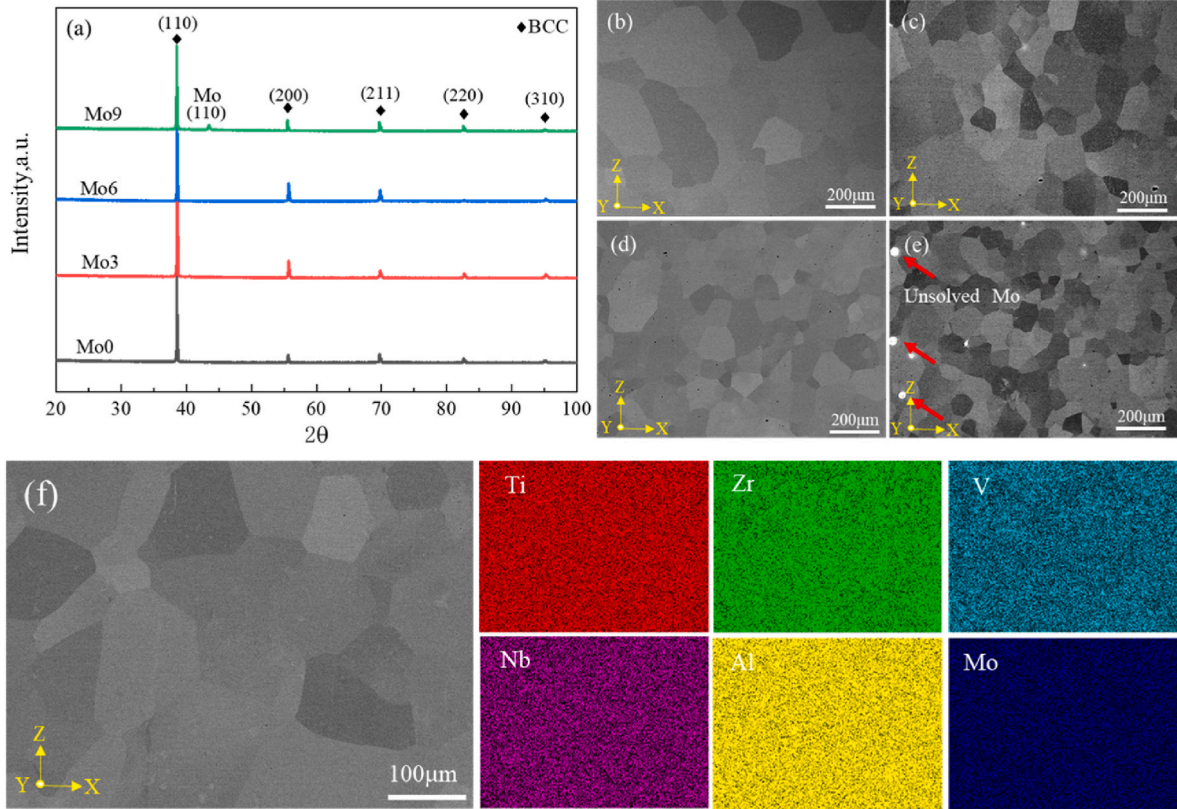


Fig. 4. (a) XRD analysis of samples with different Mo content; (b, c, d, e) BSE images of the Mo0, Mo3, Mo6, Mo9 alloys; (f) element distribution of the Mo6 alloy.

Table 1

Compositions of four RHEAs (at %).

Alloy		Ti	Zr	V	Nb	Al	Mo
Mo0	Nominal	50	18	12	15	5	0
	EDS	49.88	18.05	13.12	15.03	4.92	0
Mo3	Nominal	48.5	17.5	11.7	14.5	4.8	3
	EDS	48.53	17.45	11.71	14.53	4.72	3.06
Mo6	Nominal	47	17	11.3	14	4.7	6
	EDS	47.01	17.05	11.27	13.95	4.72	6
Mo9	Nominal	45.5	16.4	11	13.7	4.4	9
	EDS	45.76	16.63	11.15	14.01	4.44	8.01

(Fig. 8 (a, b, c)). In contrast, the Mo9 alloy exhibits an obvious brittle fracture with clear comprising river patterns and a small number of ductile dimples (Fig. 8 (d)). A comparison of the surfaces and size of the dimples among the first three alloys revealed that the Mo6 alloy exhibited smaller dimples and larger fluctuations on the sample surface. This indicates that there was no significant necking observed in the Mo6 alloy [26]. The fracture surfaces of the three alloys exhibited the formation of numerous micro-pores, which were the consequence of grain slip during the deformation process. The formation of micro-pores is indicative of a more pronounced degree of local deformation in the alloy. Additionally, the Mo6 alloy displayed numerous tearing ridges between adjacent pits, indicating a ductile tearing ridge behavior that consumes more energy. This behavior has been shown to suppress crack propagation and improve toughness, as evidenced by previous studies [58,59].

4. Discussion

4.1. Grain refinement and columnar to equiaxed transformation

The results of the EBSD analysis indicate that an increase in the Mo

content results in a gradual decrease in the mean grain sizes of the RHEAs. In accordance with the Interdependence Theory [60], grain size is primarily contingent upon the following factors: (1) ΔT_n , critical nucleation undercooling; (2) ΔT_{CS} , the amount of constitutional supercooling zone provides the nucleation undercooling; (3) X_{sd} , the distance between effective nucleation particles. It has been demonstrated that smaller ΔT_n , ΔT_{CS} , and larger ΔT_{CS} are more conducive to grain refinement [61]. The formation of the constitutional supercooling zone is primarily governed by the growth restriction factor (Q). An increase in Q value will promote more nucleation, making it easier for the alloy to achieve fine grain size during solidification [13]. In order to calculate the effect of Mo content on Q value, the curves of constitutional supercooling zone (ΔT_{CS}) with solidification fraction (f_s) for four alloys were obtained based on the non-equilibrium solidification curve (Fig. S3), as shown in Fig. 9 (a). Fig. 9 (a) illustrates that as the Mo content increases, the composition supercooling zone gradually increases. Furthermore, the Q value of the alloy was calculated in accordance with the relationship between ΔT_{CS} and Q [60], as detailed in the supporting information. The calculation of Q (Fig. 9 (b)) indicates that the Q value of the Mo9 alloy is approximately 2.7 times that of the Mo0 alloy. Consequently, as the concentration of Mo increases, the grain size decreases gradually.

According to the experimental results, the increase in Mo content not only refined the grain sizes, but also gradually transformed the equiaxed and columnar alternating microstructure into a completely fine equiaxed structure. The microstructure mainly depends on the competitive growth between the columnar growths epitaxial grown at the molting pool boundary and the equiaxed growth formed in the constitutional supercooling zone at the solid-liquid interface [23,58]. The epitaxial growth results in coarse columnar growths while the latter forms a fine equiaxed growth microstructure. The model proposed by Gäumann et al. [62] can predict this phenomenon very well:

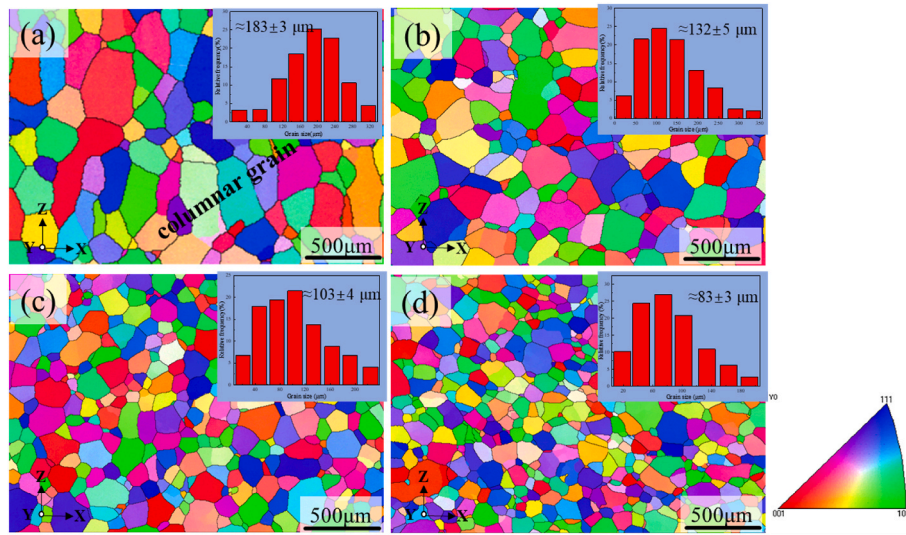


Fig. 5. EBSD IPF and grain sizes distribution with different Mo content; (a) Mo0, (b) Mo3, (c) Mo6 and (d) Mo9 alloys.

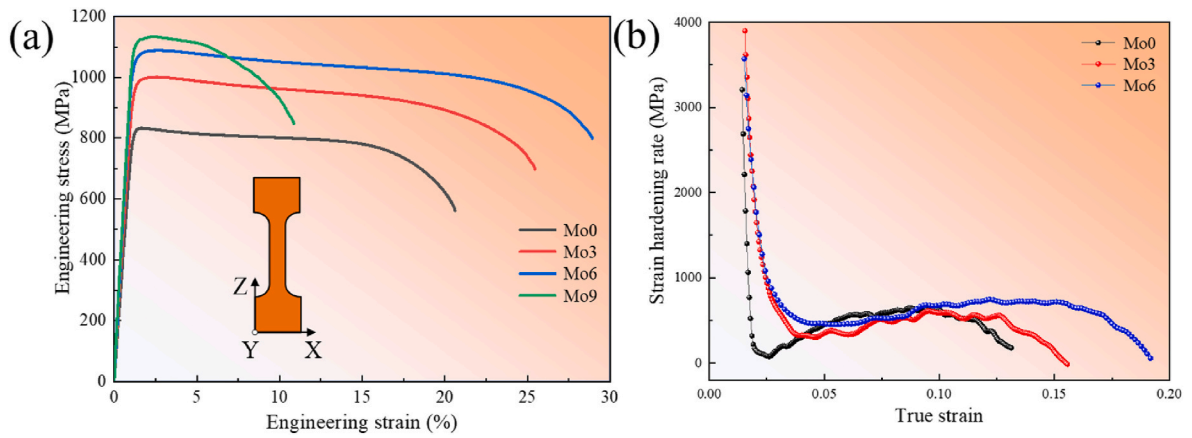


Fig. 6. Mechanical behavior of alloys with different Mo content. (a) The room-temperature tensile engineering stress-strain curves, (b) Strain-hardening rate curves of Mo0, Mo3 and Mo6 alloys.

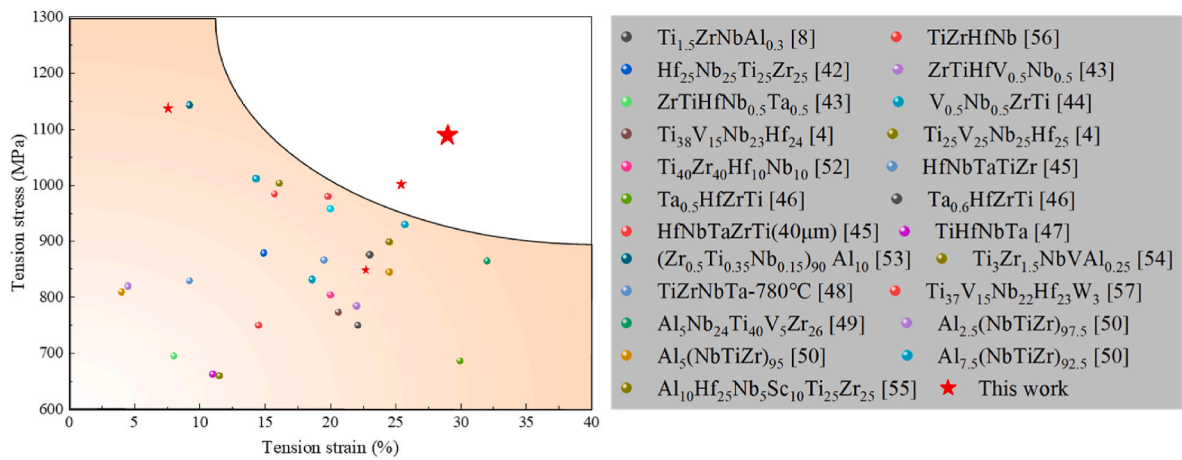


Fig. 7. Comparison of yield strength and elongation of (Ti50)100-xMox RHEAs with other typical RHEAs [4,8,42–57].

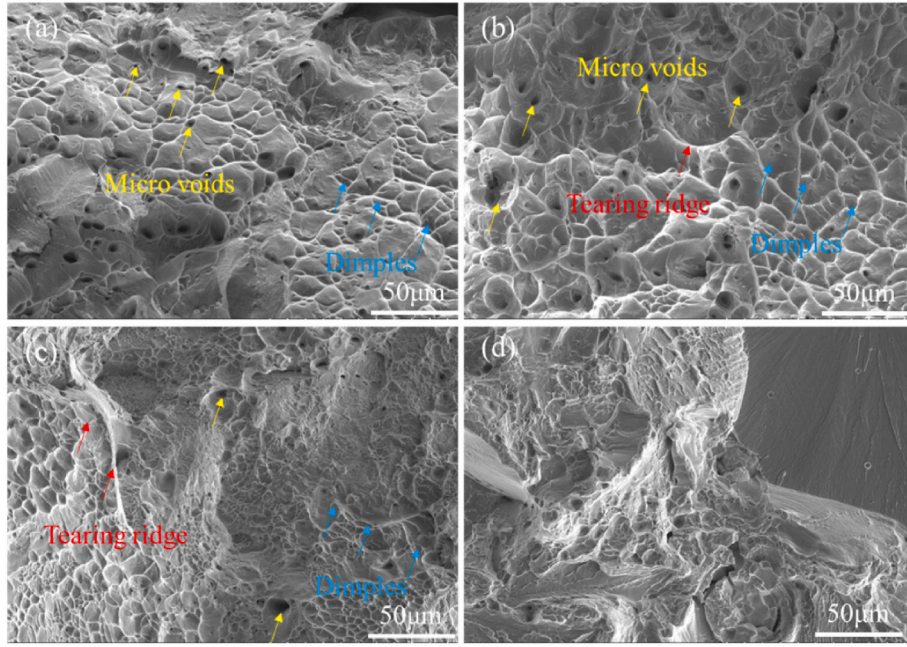


Fig. 8. The fracture surfaces of different Mo contents; (a) Mo0, (b) Mo3, (c) Mo6, (d) Mo9 alloys.

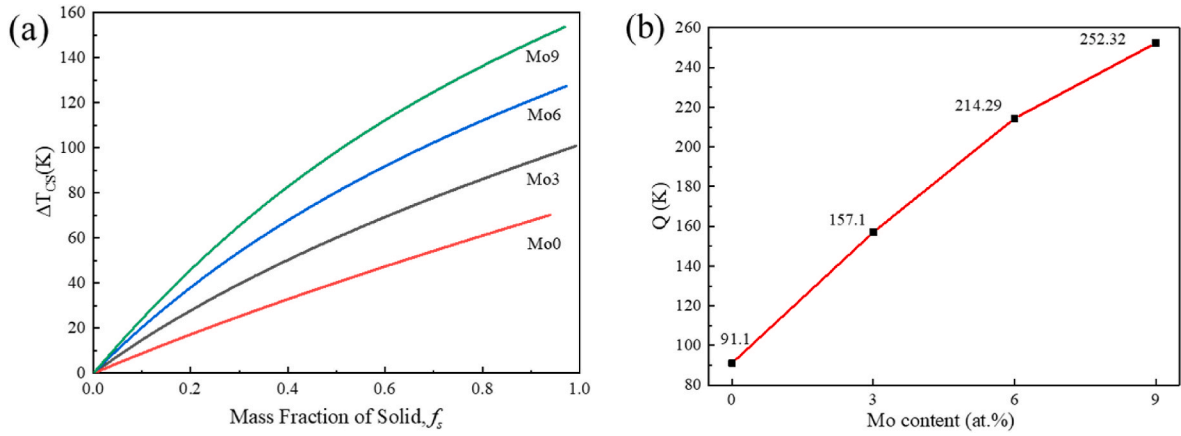


Fig. 9. (a) $\Delta T_{CS} - f_s$ curves of alloys with different Mo contents. (d) Q -values of alloys with different Mo contents.

$$G = \frac{1}{n+1} \sqrt[3]{\frac{-4\pi}{3 \ln(1-\varphi)}} N_0^{1/3} \left(1 - \frac{\Delta T_n^{n+1}}{(aR)^{(n+1)/n}}\right) (aR)^{1/n} \quad (1)$$

G represents the temperature gradient, n and a represent material-related parameters, N_0 represents the nucleation density, φ represents the volume fraction of equiaxed growths, ΔT_n represents the undercooling for nucleation, and R is the solidification rate.

According to Eq. (1), the equiaxed growth during solidification mainly depends on the temperature gradient (G), solidification rate (R), and nucleation density (N_0). Due to the same G and R of the melt pool under the same process parameters, the main factor determining the microstructure of solidification is N_0 . However, the solidification of the melt pool does not follow the simple process of pure liquid solidification, as many powders transported to the melt pool do not melt completely immediately [63,64]. Partially melted powder serves as a heterogeneous core, promoting non-uniform nucleation. As the Mo powder increases, the amount of high melting point powder in the melting pool gradually increases, resulting in an increase in heterogeneous cores in the melting pool and an increase in nucleation density [65], thereby promoting the proportion of equiaxed growth in the melting pool, as shown in Fig. 10.

Therefore, with the increase of Mo content, the alloy is more conducive to the formation of equiaxed growths.

4.2. The origin of excellent ductility

In comparison to the majority of BCC RHEAs, which sacrifice ductility to enhance yield strength, the addition of an optimal quantity of Mo in this work not only enhanced yield strength but also markedly improved ductility. In order to elucidate this anomalous phenomenon, a preliminary examination was conducted on the deformation mechanism at the grain scale. Fig. 11 (a, b, c) illustrate the three-color EBSD orientation maps of the Mo, Mo3, and Mo6 alloys following 10 % tensile deformation. The crystal structures remain in the BCC lattice, and the deformed EBSD phase maps (Fig. S4) demonstrate that no phase transformation occurs during the tensile deformation process. Consequently, the primary factors influencing the deformation mechanism are grain size and the formation of kink bands. As shown in Fig. 11 (a, b, c), lenticular structures were observed in all three alloys following deformation. The crystal disorientations of these structures are random, yet the range of crystal disorientation is approximately between 20° and 40°

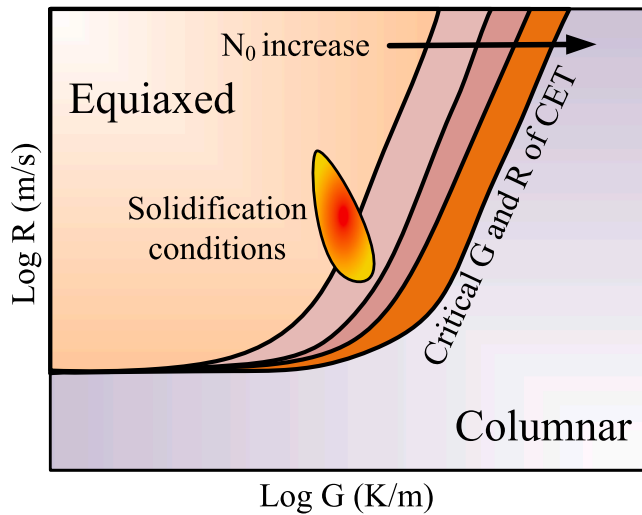


Fig. 10. Schematic diagram of columnar to equiaxed transformation and solidification conditions of Mo0, Mo3, Mo6, and Mo9 alloys.

(Fig. 11(e and f)), which precludes the possibility of mechanical twinning [66]. Consequently, these lens structures can be defined as kink bands generated during the deformation process [12,25]. The formation of lenticular structures can be attributed to stress-induced bending of the crystalline lattice during the deformation process. The coordinated deformation of a kink bands can effectively adapt to dislocation slip and alleviate stress concentration, thereby improving the ability of the alloy to undergo plastic deformation. Fig. 11 (g, h, i) illustrate the KAM maps

of three alloys following 10 % deformation. The increase in Mo content leads to a more uniform distribution of stress. This is due to the reduction in grain size, which facilitates coordinated deformation between grains, thereby reducing the degree of stress concentration and delaying the premature failure of the alloy. It can therefore be concluded that the three alloys exhibit good plastic deformation under the coordinated deformation of kink bands and grains. Furthermore, it has been demonstrated that grain refinement enhances the plastic deformation ability of the Mo6 alloy.

In order to gain further insight into the role of Mo on the deformation behavior of RHEA, TEM was employed to examine the evolution of dislocation structure and deformation mechanism in Mo0 and Mo6 alloys. The current research indicates that the deformation mechanism of RHEA is primarily based on dynamic slip bands, with the absence of twinning, stacking faults, and other mechanisms. Fig. 12 illustrates the dislocation patterns of the Mo0 and Mo6 alloys at varying stages of deformation. The diffraction condition of the TEM image is $<001>$. When deformed by about 5 %, a large number of dislocations nucleate and form numerous planar slips, accompanied by long straight screw dislocations [26,67], as shown in Fig. 12(a)–(d). With the increased strain, as shown in Fig. 12 (b) and (e), multiple planar slip formed by dislocations pile up in different directions. The slip systems in different directions inevitably generates dislocation entanglements. A large number of dislocation entanglements hinder the movement of dislocations, which is beneficial for increasing the strain hardening rate [68]. As the deformation increases, the alloy forms a mature sliding system until fracture, as shown in Fig. 12 (c). As shown in Fig. 13, due to the smaller grain size of Mo6 alloy, the grain boundary area increases. Grain boundaries have a hindering effect on dislocations (Fig. 12 (f)). When dislocations move to the grain boundaries, they are hindered by the grain boundaries, resulting in a large amount of accumulation [57]. A

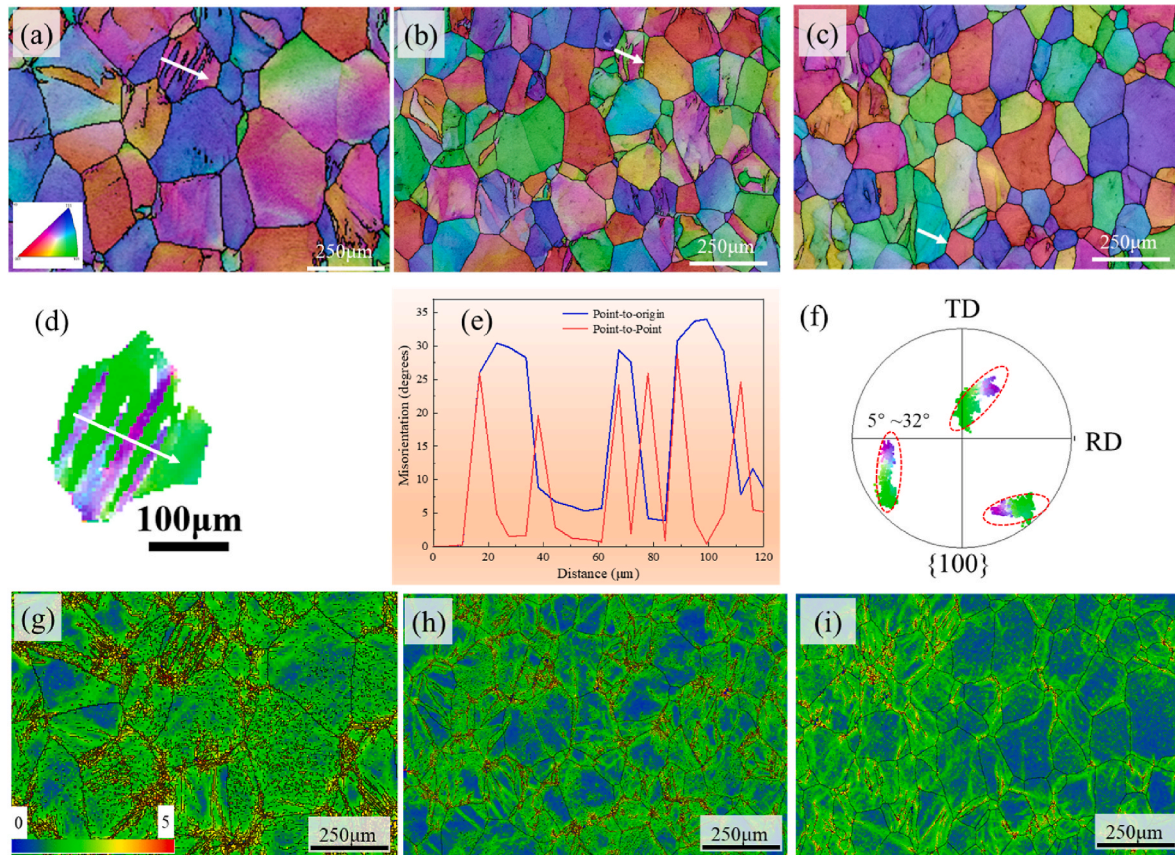


Fig. 11. EBSD analysis of 10 % deformation of three alloys: Mo0, Mo2, and Mo6; (a, b, c) IQ + IPF maps; (d, e, f) IPF maps, disorientation profiles along the kink bands and associated {100} pole map in Mo0; (g, h, i) The KAM maps.

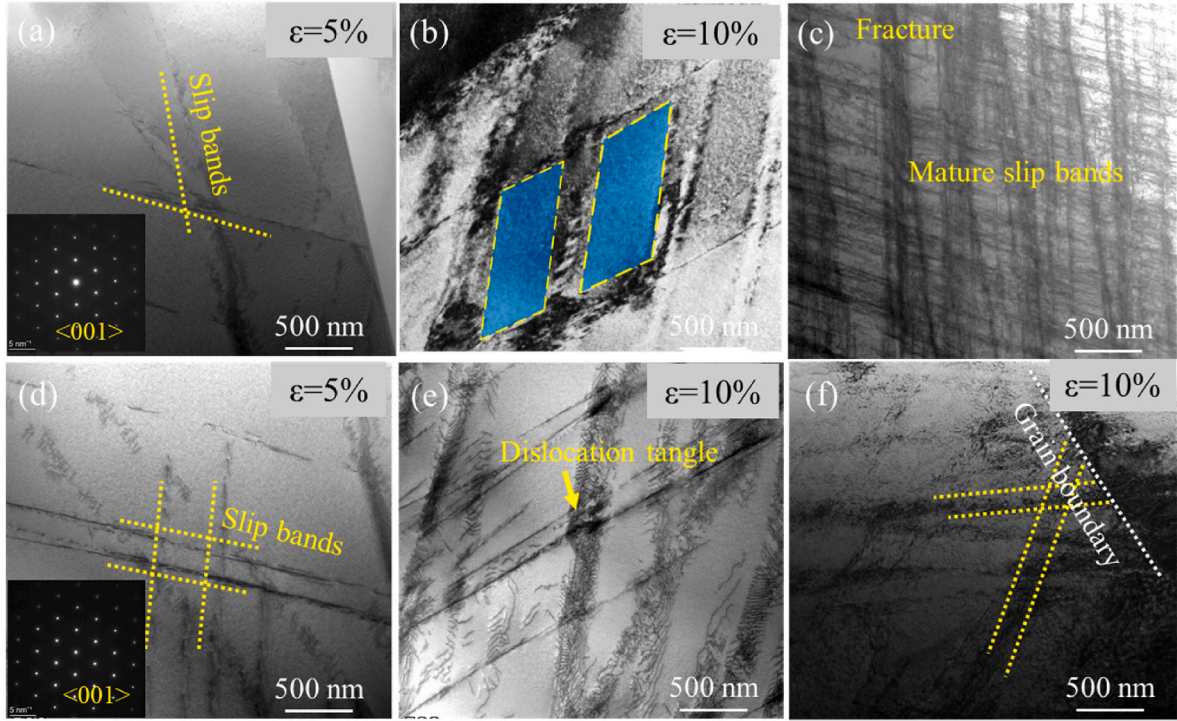


Fig. 12. TEM images of dislocation substructures of the Mo0 alloy and Mo6 alloy at different strains. (a–c) Mo0 alloy; (d–f) Mo6 alloy.

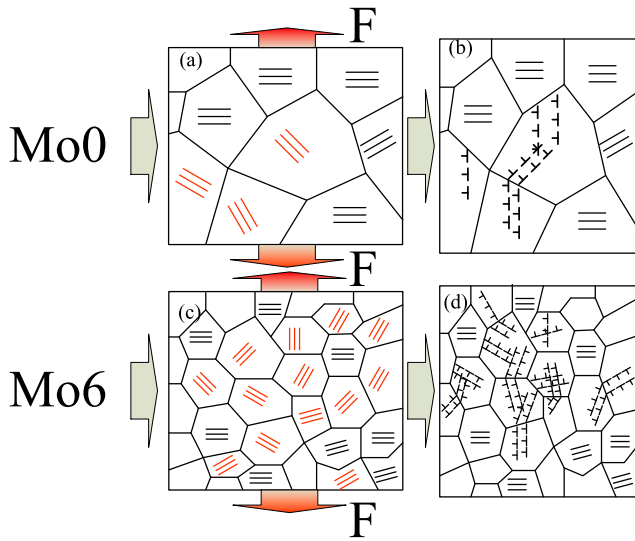


Fig. 13. Schematic of deformation process of Mo0 and Mo6 alloys.

large number of accumulated dislocations occurred near the grain boundaries, which in turn led to dislocation entanglement and cross slip [69]. As the grain size decreases, the dislocation density increases and the spacing between dislocation slip bands decreases [70], which helps to improve the ductility of the alloy. Therefore, as the Mo content increases, the grain size gradually decreases, resulting in better ductility of the alloy.

4.3. Strengthening mechanism

In order to reveal the strengthening effect caused by changes in Mo content, this section analyses the strengthening mechanism of the RHEAs. According to extant reports [18], the principal strengthening

mechanisms of the RHEAs comprise solid solution strengthening, grain refinement strengthening and dislocation strengthening. Furthermore, the presence of short-range ordering has been identified as a significant strengthening mechanism in RHEAs, with the capacity to enhance the yield strength of the RHEAs [31]. However, when the RHEAs composition change, the contribution of short-range order to the yield strength of the RHEA is much smaller than that of solid solution strengthening [71]. Furthermore, short-range order is formed through deformation and low-temperature aging, while the RHEAs have not undergone any post-treatment, so the effect of short-range order can be ignored. Consequently, the yield strength of RHEAs is predominantly determined by solid solution strengthening, grain refinement strengthening, and dislocation strengthening. The following Eq. (2) can be utilized to estimate these effects [9]:

$$\delta_Y = (\delta_{0.2})_{mix} + \Delta\delta_{ss} + \Delta\delta_G + \Delta\delta_{dis} \quad (2)$$

Where $(\delta_{0.2})_{mix}$ represents the estimated intrinsic yield strength, $\Delta\delta_{ss}$ represents the solid-solution strengthening, $\Delta\delta_G$ represents the grain boundary strengthening and $\Delta\delta_{dis}$ represents the dislocation strengthening respectively.

Grain boundary strengthening can be calculated using the Hall-Petch equation [72]:

$$\Delta\sigma_G = K_{HP}d_0^{-1/2} \quad (3)$$

Where d_0 is the average grain size, and K_{HP} is the Hall-Petch coefficient. For T50-Mox RHEAs, the K_{HP} is 240 MPa $\mu m^{1/2}$ [73]. According to the grain sizes from EBSD results, the estimated values for T50-Mox are 8.5 MPa, 9.4 MPa, 10.6 MPa and 11.8 MPa, respectively. These values manifest that the contribution of grain boundary strengthening is small.

Taylor dislocation hardening can be used to quantify the contribution of σ_{dis} by Ref. [51],

$$\delta_{dis} = M \cdot \alpha \cdot G \cdot b \cdot \sqrt{\rho} \quad (4)$$

Where M , α , G , b and ρ are the Taylor factor, empirical constant, shear modulus, Burgers vector and dislocation density, respectively. For metal

materials with BCC crystal structure, the above parameters are $M = 2.73$, $\alpha = 0.4$, $b = 0.277$ nm [1]. The shear modulus G is calculated based on the composition of four alloys, which are 41 GPa, 44 GPa, 46 GPa and 49 GPa, respectively. The dislocation densities of four alloys obtained through EBSD KAM maps are $2.4 \times 10^{11} \text{ m}^{-2}$, $2.5 \times 10^{11} \text{ m}^{-2}$, $1.9 \times 10^{11} \text{ m}^{-2}$ and $2.0 \times 10^{11} \text{ m}^{-2}$, respectively. Accordingly, the contributions of σ_{dis} in the four alloys are no more than 7 MPa, which can be reasonably ignored.

Estimated intrinsic yield strength $(\delta_{0.2})_{mix}$ can be calculated by the rule of mixture:

$$(\delta_{0.2})_{mix} = \sum c_i \sigma_{0.2i} \quad (5)$$

Where c_i represents the concentration of constituent elements, $(\sigma_{0.2})$ represents the yield strength of pure elements. The calculated values of $(\delta_{0.2})_{mix}$ for T50-Mox are 209 MPa, 215 MPa, 220 MPa and 226 MPa, respectively.

From the first two strengthening mechanisms adding Mo has little effect on either of them. It can be concluded that the primary source of yield strength enhancement for the four RHEAs is solid solution strengthening. The principal source of solid solution strengthening is the discrepancy between the shear modulus and atomic radius of the constituent elements [12,74]. Senkov-Yao [75] model is used to calculate solid solution strengthening in RHEAs at the temperature (298K):

$$\Delta\sigma_{ss} = \frac{G}{45} \left(\sum \epsilon_i^2 X_j \right)^{2/3} \quad (6)$$

Where G is the shear module of alloy, ϵ_i is the mismatch parameter.

$$\epsilon_i = (\eta_i^2 + \alpha^2 \delta_i)^{1/2} \quad (7)$$

Here, α is a deviation in the interaction foreign atoms, for the BCC solid solution, it is 16 [11]. η_i and δ_i are the average differences in the atomic modulus and atomic size, respectively.

$$\eta_i = \frac{9}{8} \sum X_j \eta_{ij} \quad (8)$$

$$\delta_i = \frac{9}{8} \sum X_j \delta_{ij} \quad (9)$$

The η_{ij} and δ_{ij} in the formula represent the differences in atomic size and shear modulus between the i th and j th atoms, respectively, which can be given as:

$$\eta_{ij} = 2(u_i - u_j) / (u_i + u_j) \quad (10)$$

$$\delta_{ij} = 2(r_i - r_j) / (r_i + r_j) \quad (11)$$

u_i and u_j represent the shear modulus of i th and j th atoms, respectively, while r_i and r_j represent the atomic radius of i th and j th atoms, respectively [12].

The parameters included in the formula are presented in Table 2 for reference [11,30]. As illustrated in Fig. 14, the solid solution strengthening model yielded results that were found to be in close agreement with the experimental outcomes, with a discrepancy of less than 10 %. In the study conducted by Xu [11] et al., it was determined that the calculated and experimental results were reliable within a range of ± 27

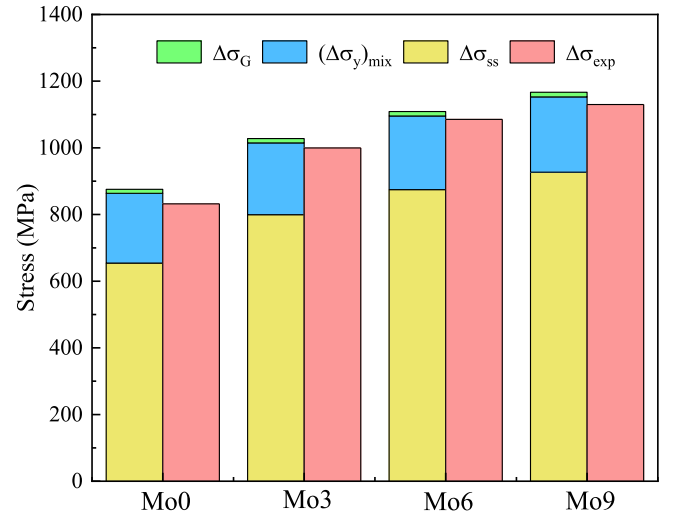


Fig. 14. Contribution of different strength mechanisms to yield strength and comparison with experimental results.

%. Therefore, based on the calculated results, approximately 80 % of the primary yield strength is due to solid solution strengthening, so the difference between the four alloys is mainly due to solid solution strengthening.

5. Conclusions

This study utilized LAM to prepare Mo-containing TiZrVNbAl-system RHEA. The main conclusions are as follows.

1. Due to the high temperature and strong convection in the melting pool. The T50-Mo RHEAs retain a single-phase BCC structure without precipitation and the high-melting-point Mo is distributed uniformly in the alloys.
2. With increasing Mo content, the constitutional supercooling zone ahead of S-L interface increases, promoting the transformation from columnar growth to equiaxed growth, and hence reducing the grain sizes.
3. The occurrence of kink behavior in RHEAs facilitates the coordinated deformation behavior of the RHEAs. Furthermore, an increase in the Mo content results in a reduction in grain size, which facilitates the coordinated deformation of the RHEAs (from 22 % to 28 %).
4. The large difference in atomic size between Mo and T50-RHEA, which achieved strong solid solution strengthening, resulting in a 34 % increase in the yield strength.

It is inevitable that the addition of Mo element in excess will result in some Mo particles remaining insoluble in the alloy. Particle strengthening is an effective method for improving yield strength of RHEAs, but there is relatively little research in this area. Therefore, it is suggested that the future research direction is to enhance RHEA with refractory element particles. The primary objective of this research is to elucidate the interface relationship between molybdenum particles and the matrix, and to ascertain the hindering effect of molybdenum on dislocations during deformation.

CRedit authorship contribution statement

Shien Liu: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Bang Dou:** Methodology, Formal analysis. **Shihai Sun:** Funding acquisition, Conceptualization. **Liang Wang:** Funding acquisition, Data curation. **Yao-Jian Liang:** Writing – review & editing, Visualization, Funding acquisition, Data curation,

Table 2

The atomic radius r_i , shear modulus G_i , and yield strength σ_{yi} of Ti, Zr, V, Nb, Al and Mo [11,30].

Elements	Ti	Zr	V	Nb	Al	Mo
$r_i(\text{pm})$	142.5	156.3	132.9	142.9	135.9	139
$G_i(\text{GPa})$	44	33	47	38	26	126
$\sigma_{yi}(\text{MPa})$	195	35	103	240	276	420

Conceptualization. **Yunfei Xue**: Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msea.2025.148042>.

Data availability

Data will be made available on request.

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